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"Relativity theory is so romantic. And so sad."
- Makise Kurisu

PROOF EXERCISES

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Wisam Yassar Mahardika

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1 SENTENTIAL LOGIC

1.1 Deductive Reasoning and Logical Connectives

1.1.1 Exercises:

1. Analyze the logical forms of the following statements:
   (a) We'll have either a reading assignment or a homework problems, but we won't have both homework problems and a test.
   (b) You won't go skiing or you will and there won't be any snow.
   (c) $\sqrt{7} \nleq 2$.
2. Analyze the logical forms of the following statements:
   (a) Either John and Bill are both telling the truth, or neither of them is.
   (b) I'll have either fish or chicken, but I won't have both fish and mashed potatoes.
   (c) 3 is a common divisor of 6, 9 and 15.
3. Analyze the logical forms of the following statements
   (a) Alice and Bob are not both in the room.
   (b) Alice and Bob are both not in the room.
   (c) Either Alice or Bob is not in the room.
   (d) Neither Alice nor Bob is in the room.
4. Which of the following expressions are well-formed formulas?
   (a) $\neg(\neg P \vee \neg\neg R)$.
   (b) $\neg(P, Q, \wedge R)$.
   (c) $P \wedge \neg P$.
   (d) $(P \wedge Q)(P \vee R)$.
5. Let P stand for the statement "I will buy the pants" and S for the statement "I will buy the shirt." What English sentences are represented by the following expressions?
   (a) $\neg(P \wedge \neg S)$.
   (b) $\neg P \wedge \neg S$.
   (c) $\neg P \vee \neg S$.
6. Let S stand for the statement "Steve is happy" and G for "George is happy." What English sentences are represented by the following expressions?
   (a) $(S \vee G) \wedge (\neg S \vee \neg G)$.
   (b) $[S \vee (G \wedge \neg S)] \vee \neg G$.
   (c) $S \vee [G \wedge (\neg s \vee \neg G)]$.
7. Identify the premises and conclusions of the following deductive arguments and analyze their logical forms. Do you think the reasoning is valid? (Although you will have only your intuition to guide you in answering this last question, in the next section we will develop some techniques for determining the validity of arguments.)

(a) Jane and Pete won't both win the math prize. Pete will win either the math prize or the chemistry prize. Jane will win the math prize. Therefore, Pete will win the chemistry prize.

(b) The main course will be either beef or fish. The vegetable will be either peas or corn. We will not have both fish as a main course and corn as a vegetable. Therefore, we will not have both beef as a main course and peas as a vegetable.

(c) Either John or Bill is telling the truth. Either Sam or Bill is lying. Therefore, either John is telling the truth or Sam is lying.

(d) Either sales will go up and the boss will be happy, or expenses will go up and the boss won't be happy. Therefore, sales and expenses will not both go up.

1.1.2 Solutions:

1. Logical Forms:

   (a) **Statement:** We'll have either a reading assignment or a homework problem, but we won't have both homework problems and a test.
   Atomic Statements: Let R denote "We'll have a reading assignment," H denote "We'll have homework problems," and T denote "We'll have a test."
   Logical Form: $(R \vee H) \wedge \neg(H \wedge T)$
   Reasoning: "Either R or H" is represented by $(R \vee H)$. The connective "but" functions as a logical conjunction ($\wedge$). "Not both H and T" translates to $\neg(H \wedge T)$.

   (b) **Statement:** You won't go skiing or you will and there won't be any snow.
   Atomic Statements: Let S denote "You will go skiing," and N denote "There will be snow."
   Logical Form: $\neg S \vee (S \wedge \neg N)$
   Reasoning: "You won't go skiing" is $\neg S$. The main connective is "or" ($\vee$). The second clause "you will go skiing and there won't be any snow" combines S and $\neg N$ with an "and" ($\wedge$).

   (c) **Statement:** $\sqrt{7} \nleq 2$.
   Atomic Statement: Let P denote $\sqrt{7} \leq 2$.
   Logical Form: $\neg P$
   Reasoning: The symbol $\nleq$ directly negates the relation $\leq$.

2. Logical Forms:

   (a) **Statement:** Either John and Bill are both telling the truth, or neither of them is.
   Atomic Statements: Let J denote "John is telling the truth," and B denote "Bill is telling the truth."
   Logical Form: $(J \wedge B) \vee (\neg J \wedge \neg B)$
   Reasoning: "Both telling the truth" is $(J \wedge B)$. "Neither is telling the truth" means John is not telling the truth and Bill is not telling the truth $(\neg J \wedge \neg B)$. These two possibilities are joined by "either ... or" ($\vee$).

   (b) **Statement:** I'll have either fish or chicken, but I won't have both fish and mashed potatoes.
   Atomic Statements: Let F denote "I'll have fish," C denote "I'll have chicken," and M denote "I'll have mashed potatoes."
   Logical Form: $(F \vee C) \wedge \neg(F \wedge M)$
   Reasoning: "Either fish or chicken" if $(F \vee C)$. "But" represents conjunction ($\wedge$). "Not both fish and mashed potatoes" represents $\neg(F \wedge M)$.

(c) **Statement:** 3 is a common divisor of 6, 9 and 15.
Atomic Statements: Let D_1 denote "3 is a divisor of 6," D_2 denote "3 is a divisor of 9," and D_3 denote "3 is a divisor of 15."
Logical Form: $D_1 \wedge D_2 \wedge D_3$
Reasoning: Stating that 3 is a common divisor means 3 divides 6 *and* 3 divides 9 *and* 3 divides 15.

3. Comparing Logical Forms:
Let A stand for "Alice is in the room" and B stand for "Bob is in the room."

(a) **Statement:** Alice and Bob are not both in the room.
Logical Form: $\neg(A \wedge B)$
Reasoning: Negates the conjunction that both Alice and Bob are in the room.

(b) **Statement:** Alice and Bob are both not in the room.
Logical Form: $\neg A \wedge \neg B$
Reasoning: Asserts that Alice is not in the room *and* Bob is not in the room.

(c) **Statement:** Either Alice or Bob is not in the room.
Logical Form: $\neg A \vee \neg B$
Reasoning: Asserts that at least one of them is absent.

(d) **Statement:** Neither Alice nor Bob is in the room.
Logical Form: $\neg(A \vee B)$
Reasoning: Denies that either Alice or Bob is in the room.

4. Well Formed Formulas:

(a) **Formula:** $\neg(\neg P \vee \neg\neg R)$
Answer: Yes, Well-Formed Formula (WFF).
Reasoning: P and R are atomic formulas. $\neg P$ and $\neg\neg R$ are well-formed, connected properly by $\vee$, and overall negated by $\neg$.

(b) **Formula:** $\neg(P, Q, \wedge R)$
Answer: No, not a WFF.
Reasoning: Commas are not logical connectives, and the binary operator $\wedge$ lacks a let operand inside the comma list.

(c) **Formula:** $P \wedge \neg P$
Answer: Yes, Well-Formed Formula (WFF).
Reasoning: Standard binary conjunction connecting atomic formula P and well-formed formula $\neg P$.

(d) **Formula:** $(P \wedge Q)(P \vee R)$
Answer: No, not a WFF.
Reasoning: Two parenthesized formulas are juxtaposed without a binary logical operator connecting them.

5. English Translations:
Let P stand for "I will buy the pants" and S for "I will buy the shirt."

(a) **Expression:** $\neg(P \wedge \neg S)$
Literal Translation: It is not the case that i will buy the pants and not buy the shirt.
Natural English: I won't buy the pants without buying the shirt.

(b) **Expression:** $\neg P \wedge \neg S$
Literal Translation: I will not buy the pants and i will not buy the shirt.
Natural English: I will buy neither the pants nor the shirt.

(c) **Expression:** $\neg P \vee \neg S$
Literal Translation: I will not buy the pants or i will not buy the shirt.
Natural English: I won't buy both the pants and the shirt.

6. English Translation:
Let S stand for "Steve is happy" and G for "George is happy."

(a) **Expression:** $(S \vee G) \wedge (\neg S \vee \neg G)$
Translation: Either Steve or George is happy, but they are not both happy.

(b) **Expression:** $[S \vee (G \wedge \neg S)] \vee \neg G$
Translation: Steve is happy, or George is happy and Steve is not happy, or George is not happy.

(c) **Expression:** $S \vee [G \wedge (\neg S \vee \neg G)]$
Translation: Steve is happy, or George is happy and either Steve or George is not happy.

7. Argument Structure & Validity Analysis

(a) **Premises:** $\neg(J_M \wedge P_M)$, $P_M \vee P_C$, J_M
Conclusion: P_C
Validity: Valid.
Reasoning: Since J_M is True, $\neg(J_M \wedge P_M)$ implies P_M must be False. Since $P_M \vee P_C$ is True and P_M is False, P_C must be true.

(b) **Premises:** $B \vee F$, $P \vee C$, $\neg(F \wedge C)$
Conclusion: $\neg(B \wedge P)$
Validity: Invalid.
Reasoning: Consider $B = \text{True}, F = \text{False}, P = \text{True}, C = \text{False}$. All premises are true ($B \vee F$ is True, $P \vee C$ is True, $\neg(F \wedge C)$ is True), but the conclusion $\neg(B \wedge P)$ is False because both B and P are True.

(c) **Premises:** $J \vee B$, $\neg S \vee \neg B$
Conclusion: $J \vee \neg S$
Validity: Valid.
Reasoning: If B is False, then J must be True from $J \vee B$. If B is True, then $\neg S$ must be True from $\neg S \vee \neg B$. In either case, $J \vee \neg S$ is True.

(d) **Premises:** $(S \wedge H) \vee (E \wedge \neg H)$
Conclusion: $\neg(S \wedge E)$
Validity: Invalid.
Reasoning: Consider $S = \text{True}, E = \text{True}, H = \text{True}$. The premise $(S \wedge H) \vee (E \wedge \neg H)$ is True because $S \wedge H$ is True. However, both S and E are True, so $S \wedge E$ is True, making the conclusion $\neg(S \wedge E)$ False.

1.2 Truth Tables

1.2.1 Exercises:

1. Make truth tables for the following formulas:
   (a) $\neg P \vee Q$.
   (b) $(S \vee G) \wedge (\neg S \vee \neg G)$.
2. Make truth tables for the following formulas:
   (a) $\neg[P \wedge (Q \vee \neg P)]$.
   (b) $(P \vee Q) \wedge (\neg P \vee R)$.
3. In this exercise we will use the symbol $+$ to mean *exclusive or*. In other words, $P + Q$ means "P or Q, but not both."
   (a) Make a truth table for $P + Q$.
   (b) Find a formula using only the connectives $\wedge$, $\vee$, and $\neg$ that is equivalent to $P + Q$. Justify your answer with a truth table.
4. Find a formula using only the connectives $\wedge$ and $\neg$ that is equivalent to $P \vee Q$. Justify your answer with a truth table.
5. Some mathematicians use the symbol $\downarrow$ to mean *nor*. In other words $P \downarrow Q$ means "neither P nor Q."
   (a) Make a truth table for $P \downarrow Q$.
   (b) Find a formula using only the connectives $\wedge$, $\vee$, and $\neg$ that is equivalent to $P \downarrow Q$.
   (c) Find formulas using only the connective $\downarrow$ that are equivalent to $\neg P$, $P \vee Q$, and $P \wedge Q$.
6. Some mathematicians write $P \mid Q$ to mean "P and Q are not both true." (This connective is called *nand*, and is used in the study of circuits in computer science.)
   (a) Make a truth table for $P \mid Q$.
   (b) Find a formula using only the connective $\wedge$, $\vee$, and $\neg$ that is equivalent to $P \mid Q$.
   (c) Find formulas using only the connective $\mid$ that are equivalent to $\neg P$, $P \vee Q$, and $P \wedge Q$.
7. Use truth tables to determine whether or not the arguments in exercise 7 of Section 1.1 are valid.
8. Use truth tables to determine which of the following formulas are equivalent to each other:
   (a) $(P \wedge Q) \vee (\neg P \wedge \neg Q)$.
   (b) $\neg P \wedge Q$.
   (c) $(P \vee \neg Q) \wedge (Q \vee \neg P)$.
   (d) $\neg(P \vee Q)$.
   (e) $(Q \wedge P) \vee \neg P$.
9. Use truth tables to determine which of these statements are tautologies, which are contradictions, and which are neither:
   (a) $(P \vee Q) \wedge (\neg P \vee \neg Q)$.

(b) $(P \vee Q) \wedge (\neg P \wedge \neg Q)$.

(c) $(P \vee Q) \vee (\neg P \vee \neg Q)$.

(d) $[P \wedge (Q \vee \neg R)] \vee (\neg P \vee R)$.

10. Use truth tables to check these laws:

    (a) The second DeMorgan's law. (The first was checked in the text.)

    (b) The distributive laws.

11. Use the laws stated in the text to find simpler formulas equivalent to these formulas. (See Examples 1.2.5 and 1.2.7.)

    (a) $\neg(\neg P \wedge \neg Q)$.

    (b) $(P \wedge Q) \vee (P \wedge \neg Q)$.

    (c) $\neg(P \wedge \neg Q) \vee (\neg P \wedge Q)$.

12. Use the laws stated in the text to find simpler formulas equivalent to these formulas. (See examples 1.2.5 and 1.2.7.)

    (a) $\neg(\neg P \vee Q) \vee (P \wedge \neg R)$.

    (b) $\neg(\neg P \wedge Q) \vee (P \vee \neg R)$.

    (c) $(P \wedge R) \vee [\neg R \wedge (P \vee Q)]$.

13. Use the first DeMorgan's law and the double negation law to derive the second DeMorgan's law.

14. Note that the associative laws say only that parentheses are unecessary when combining *three* statements with $\wedge$ or $\vee$. In fact, these laws can be used to justify leaving parentheses out when more than three statements are combined. Use associative laws to show that $[P \wedge (Q \wedge R)] \wedge S$ is equivalent to $(P \wedge Q) \wedge (R \wedge S)$.

15. How many lines will there be in the truth table for a statement containing n letters?

16. Find a formula involving the connectives $\wedge$, $\vee$, and $\neg$ that has the following truth table:

P	Q	???
F	F	T
F	T	F
T	F	T
T	T	T

17. Find a formula involving the connectives $\wedge$, $\vee$, and $\neg$ that has the following truth table:

P	Q	???
F	F	F
F	T	T
T	F	T
T	T	F

18. Suppose the conclusion of an argument is a tautology. What can you conclude about the validity of the argument? What if the conclusion is a contradiction? What if one of the premises is either a tautology or a contradiction?

1.2.2 Solutions:

1. Truth Tables:

   (a) **Formula:** $\neg P \vee Q$.
   Truth Table:

P	Q	$\neg P$	$\neg P \vee Q$
T	T	F	T
T	F	F	F
F	T	T	T
F	F	T	T

   (b) **Formula:** $(S \vee G) \wedge (\neg S \vee \neg G)$.
   Truth Table:

S	G	$S \vee G$	$\neg S \vee \neg G$	$(S \vee G) \wedge (\neg S \vee \neg G)$
T	T	T	F	F
T	F	T	T	T
F	T	T	T	T
F	F	F	T	F

2. Truth Tables:

   (a) **Formula:** $\neg[P \wedge (Q \vee \neg P)]$.
   Truth Table:

P	Q	$\neg P$	$Q \vee \neg P$	$P \wedge (Q \vee \neg P)$	$\neg[P \wedge (Q \vee \neg P)]$
T	T	F	T	T	F
T	F	F	F	F	T
F	T	T	T	F	T
F	F	T	T	F	T

   (b) **Formula:** $(P \vee Q) \wedge (\neg P \vee R)$.
   Truth Table:

P	Q	R	$P \vee Q$	$\neg P \vee R$	$(P \vee Q) \wedge (\neg P \vee R)$
T	T	T	T	T	T
T	T	F	T	F	F
T	F	T	T	T	T
T	F	F	T	F	F
F	T	T	T	T	T
F	T	F	T	T	T
F	F	T	F	T	F
F	F	F	F	T	F

3. Exclusive Or ($+$):

   (a) **Truth Table for** $P + Q$:

P	Q	$P + Q$
T	T	F
T	F	T
F	T	T
F	F	F

(b) **Equivalent Formula:** $(P \vee Q) \wedge \neg(P \wedge Q)$.
Reasoning: "P or Q" translates to $(P \vee Q)$, and "but not both" translates to $\wedge\neg(P \wedge Q)$.

4. Equivalent to $P \vee Q$ Using Only $\wedge$ and $\neg$:
Formula: $\neg(\neg P \wedge \neg Q)$.
Truth Table:

P	Q	$P \vee Q$	$\neg(\neg P \wedge \neg Q)$
T	T	T	T
T	F	T	T
F	T	T	T
F	F	F	F

Reasoning: By DeMorgan's laws, negating the conjunction of two negated variables yields the disjunction of the original variables.

5. Nor $(P \downarrow Q)$:

(a) **Truth Table for** $P \downarrow Q$:

P	Q	$P \downarrow Q$
T	T	F
T	F	F
F	T	F
F	F	T

(b) **Equivalent Formula:** $\neg(P \vee Q)$.

(c) **Equivalent Formulas Using Only** $\downarrow$:

- $\neg P \equiv P \downarrow P$.
- $P \vee Q \equiv (P \downarrow Q) \downarrow (P \downarrow Q)$.
- $P \wedge Q \equiv (P \downarrow P) \downarrow (Q \downarrow Q)$.

Reasoning: "Neither P nor Q" is the exact negation of "P or Q".

6. Nand $(P \mid Q)$:

(a) **Truth Table for** $P \mid Q$:

P	Q	$P \mid Q$
T	T	F
T	F	T
F	T	T
F	F	T

(b) **Equivalent Formula:** $\neg(P \wedge Q)$.

(c) **Equivalent Formulas Using Only** $\mid$:

- $\neg P \equiv P \mid P$.
- $P \vee Q \equiv (P \mid P) \mid (Q \mid Q)$.
- $P \wedge Q \equiv (P \mid Q) \mid (P \mid Q)$.

Reasoning: "Not both true" is the direct negation of the conjunction.

7. Validity Analysis via Truth Tables:

(a) Argument (a): **Premises:** $\neg(J_M \wedge P_M), P_M \vee P_C, J_M$.
Conclusion: P_C.
Truth Table:

J_M	P_M	P_C	$J_M \wedge P_M$	$\neg(J_M \wedge P_M)$	$P_M \vee P_C$	J_M	P_C
T	T	T	T	F	T	T	T
T	T	F	T	F	T	T	F
T	F	T	F	T	T	T	T
T	F	F	F	T	F	T	F
F	T	T	F	T	T	F	T
F	T	F	F	T	T	F	F
F	F	T	F	T	T	F	T
F	F	F	F	T	F	F	F

Result: *Valid*. In the only row where all three premises are True (Row 3). the conclusion (P_C) is also True.

(b) Argument (b):
Premises: $B \vee F, P \vee C, \neg(F \wedge C)$.
Conclusion: $\neg(B \wedge P)$.
Truth Table:

B	F	P	C	$B \vee F$	$P \vee C$	$\neg(F \wedge C)$	$\neg(B \wedge P)$
T	T	T	T	T	T	F	F
T	T	T	F	T	T	T	F
T	T	F	T	T	T	F	T
T	T	F	F	T	F	T	T
T	F	T	T	T	T	T	F
T	F	T	F	T	T	T	F
T	F	F	T	T	T	T	T
T	F	F	F	T	F	T	T
F	T	T	T	T	T	F	T
F	T	T	F	T	T	T	T
F	T	F	T	T	T	F	T
F	T	F	F	T	F	T	T
F	F	T	T	F	T	T	T
F	F	T	F	F	T	T	T
F	F	F	T	F	T	T	T
F	F	F	F	F	F	T	T

Result: *Invalid*. In the 2nd, 5th, and 6th rows, all three premises are True, but the conclusion is False.

(c) Argument (c):
Premises: $J \vee B, \neg S \vee \neg B$.
Conclusion: $J \vee \neg S$.
Truth Table:

J	B	S	$\neg B$	$\neg S$	$J \vee B$	$\neg S \vee \neg B$	$J \vee \neg S$
T	T	T	F	F	T	F	T
T	T	F	F	T	T	T	T
T	F	T	T	F	T	T	T
T	F	F	T	T	T	T	T
F	T	T	F	F	T	F	F
F	T	F	F	T	T	T	T
F	F	T	T	F	F	T	F
F	F	F	T	T	F	T	T

Result: *Valid*. In every row where both premises are True (Rows 2, 3, 4, and 6), the conclusion is also True.

(d) Argument (d):
Premises: $(S \wedge H) \vee (E \wedge \neg H)$.
Conclusion: $\neg(S \wedge E)$.
Truth Table:

S	E	H	$S \wedge H$	$E \wedge \neg H$	$(S \wedge H) \vee (E \wedge \neg H)$	$S \wedge E$	$\neg(S \wedge E)$
T	T	T	T	F	T	T	F
T	T	F	F	T	T	T	F
T	F	T	T	F	T	F	T
T	F	F	F	F	F	F	T
F	T	T	F	F	F	F	T
F	T	F	F	T	T	F	T
F	F	T	F	F	F	F	T
F	F	F	F	F	F	F	T

Result: *Invalid*. In the 1st and 2nd rows, the premise is True, but the conclusion is False.

8. Equivalence Analysis via Truth Tables:
Truth Table:

P	Q	$(P \wedge Q) \vee (\neg P \wedge \neg Q)$	$\neg P \wedge Q$	$(P \vee \neg Q) \wedge (Q \vee \neg P)$	$\neg(P \vee Q)$	$(Q \wedge P) \vee \neg P$
T	T	T	F	T	F	T
T	F	F	F	F	F	F
F	T	F	T	F	F	T
F	F	T	F	T	T	T

Result: The colums $(P \wedge Q) \vee (\neg P \wedge \neg Q)$ and $(P \vee \neg Q) \wedge (Q \vee \neg P)$ have the exact same truth values in every row, Therefore $(P \wedge Q) \vee (\neg P \wedge \neg Q) \equiv (P \vee \neg Q) \wedge (Q \vee \neg P)$.

9. Tautology, Contradiction, or Neither:

(a) $(P \vee Q) \wedge (\neg P \vee \neg Q)$

(b) $(P \vee Q) \wedge (\neg P \wedge \neg Q)$

(c) $(P \vee Q) \vee (\neg P \vee \neg Q)$

(d) $[P \wedge (Q \vee \neg R)] \vee (\neg P \vee R)$

Truth Table:

P	Q	R	(a)	(b)	(c)	(d)
T	T	T	F	F	T	T
T	T	F	F	F	T	T
T	F	T	T	F	T	T
T	F	F	T	F	T	T
F	T	T	T	F	T	T
F	T	F	T	F	T	T
F	F	T	F	F	T	T
F	F	F	F	F	T	T

Result:

- (a) is *Neither* (True in some rows, False in others).
- (b) is a *Contradiction* (False in all rows).
- (c) is a *Tautology* (True in all rows).
- (d) is a *Tautology* (True in all rows).

10. Checking Laws:

(a) **The second DeMorgan's law:** $(\neg(P \wedge Q) \equiv \neg P \vee \neg Q)$

P	Q	$P \wedge Q$	$\neg(P \wedge Q)$	$\neg P$	$\neg Q$	$\neg P \vee \neg Q$
T	T	T	**F**	F	F	**F**
T	F	F	**T**	F	T	**T**
F	T	F	**T**	T	F	**T**
F	F	F	**T**	T	T	**T**

Result: The column values for $\neg(P \wedge Q)$ and $\neg P \vee \neg Q$ match perfectly, proving the law.

(b) **The Distributive Laws**
Law 1: $P \wedge (Q \vee R) \equiv (P \wedge Q) \vee (P \wedge R)$

P	Q	R	$Q \vee R$	$P \wedge (Q \vee R)$	$P \wedge Q$	$P \wedge R$	$(P \wedge Q) \vee (P \wedge R)$
T	T	T	T	**T**	T	T	**T**
T	T	F	T	**T**	T	F	**T**
T	F	T	T	**T**	F	T	**T**
T	F	F	F	**F**	F	F	**F**
F	T	T	T	**F**	F	F	**F**
F	T	F	T	**F**	F	F	**F**
F	F	T	T	**F**	F	F	**F**
F	F	F	F	**F**	F	F	**F**

Result: The values match perfectly, proving the first distributive law.
Law 2: $P \vee (Q \wedge R) \equiv (P \vee Q) \wedge (P \vee R)$

P	Q	R	$Q \wedge R$	$P \vee (Q \wedge R)$	$P \vee Q$	$P \vee R$	$(P \vee Q) \wedge (P \vee R)$
T	T	T	T	**T**	T	T	**T**
T	T	F	F	**T**	T	T	**T**
T	F	T	F	**T**	T	T	**T**
T	F	F	F	**T**	T	T	**T**
F	T	T	T	**T**	T	T	**T**
F	T	F	F	**F**	T	F	**F**
F	F	T	F	**F**	F	T	**F**
F	F	F	F	**F**	F	F	**F**

Result: The values match perfectly, proving the second distributive law.

11. Simpler Formulas:

(a) $\neg(\neg P \wedge \neg Q)$

$$\begin{aligned}\neg(\neg P \wedge \neg Q) &\equiv \neg\neg P \vee \neg\neg Q \quad \text{DeMorgan's law}\\ &\equiv P \vee Q \quad \text{double negation law}\end{aligned}$$

(b) $(P \wedge Q) \vee (P \wedge \neg Q)$

$$\begin{aligned}(P \wedge Q) \vee (P \wedge \neg Q) &\equiv P \wedge (Q \vee \neg Q) \quad \text{Distributive law}\\ &\equiv P \wedge \mathit{True} \quad \text{tautology law}\\ &\equiv P \quad \text{identity law}\end{aligned}$$

(c) $\neg(P \wedge \neg Q) \vee (\neg P \wedge Q)$

$$\begin{aligned}\neg(P \wedge \neg Q) \vee (\neg P \wedge Q) &\equiv (\neg P \vee \neg\neg Q) \vee (\neg P \wedge Q) \quad \text{DeMorgan's law}\\ &\equiv (\neg P \vee Q) \vee (\neg P \wedge Q) \quad \text{double negation law}\\ &\equiv \neg P \vee (Q \vee (\neg P \wedge Q)) \quad \text{associative law}\\ &\equiv \neg P \vee ((Q \vee \neg P) \wedge (Q \vee Q)) \quad \text{distributive law}\\ &\equiv \neg P \vee ((Q \vee \neg P) \wedge Q) \quad \text{idempotent law}\\ &\equiv \neg P \vee Q \quad \text{absorption law}\end{aligned}$$

12. Simpler Formulas:

(a) $\neg(\neg P \vee Q) \vee (P \wedge \neg R)$

$$\begin{aligned}\neg(\neg P \vee Q) \vee (P \wedge \neg R) &\equiv (\neg\neg P \wedge \neg Q) \vee (P \vee \neg R) \quad \text{DeMorgan's law}\\ &\equiv (P \wedge \neg Q) \vee (P \wedge \neg R) \quad \text{double negation law}\\ &\equiv P \wedge (\neg Q \vee \neg R) \quad \text{distributive law}\\ &\equiv P \wedge \neg(Q \wedge R) \quad \text{DeMorgan's law}\end{aligned}$$

(b) $\neg(\neg P \wedge Q) \vee (P \vee \neg R)$

$$\begin{aligned}\neg(\neg P \wedge Q) \vee (P \vee \neg R) &\equiv (\neg\neg P \vee \neg Q) \vee (P \vee \neg R) \quad \text{DeMorgan's law}\\ &\equiv (P \vee \neg Q) \vee (P \vee \neg R) \quad \text{double negation law}\\ &\equiv (P \vee P) \vee (\neg Q \vee \neg R) \quad \text{associative and commutative laws}\\ &\equiv P \vee (\neg Q \vee \neg R) \quad \text{idempotent law}\\ &\equiv P \vee \neg(Q \wedge R) \quad \text{DeMorgan's law}\end{aligned}$$

(c) $(P \wedge R) \vee [\neg R \wedge (P \vee Q)]$

$$\begin{aligned}(P \wedge R) \vee [\neg R \wedge (P \vee Q) &\equiv (P \wedge R) \vee [(\neg R \wedge P) \vee (\neg R \wedge Q)] \quad \text{distributive law}\\ &\equiv [(P \wedge R) \vee (\neg R \wedge P)] \vee (\neg R \wedge Q) \quad \text{associative law}\\ &\equiv [(R \wedge P) \vee (\neg R \wedge P)] \vee (\neg R \wedge Q) \quad \text{commutative law}\\ &\equiv [(R \vee \neg R) \wedge P] \vee (\neg R \wedge Q) \quad \text{distributive law}\\ &\equiv [\mathit{True} \wedge P] \vee (\neg R \wedge Q) \quad \text{tautology law}\\ &\equiv P \vee (\neg R \wedge Q) \quad \text{identity law}\end{aligned}$$

13. **Deriving the Second DeMorgan's Law:**
Start with the negation of a conjunction, $\neg(P \wedge Q)$.
By the double negation law ($P \equiv \neg\neg P$), we can rewrite the inner variables P and Q:

$$\neg(P \wedge Q) \equiv \neg(\neg\neg P \wedge \neg\neg Q)$$

Applying the first DeMorgan's law, $\neg(A \wedge B) \equiv \neg A \vee \neg B$, to the terms inside the outer negation:

$$\neg(\neg\neg P \wedge \neg\neg Q) \equiv \neg\neg(\neg P \vee \neg Q)$$

Finally, applying the double negation law again removes the two outermost negations:

$$\neg\neg(\neg P \vee \neg Q) \equiv \neg P \vee \neg Q$$

Therefore, $\neg(P \wedge Q) \equiv \neg P \vee \neg Q$.

14. **Associative Laws for Multiple Statements:**
By the associative law for conjunction, $A \wedge (B \wedge C) \equiv (A \wedge B) \wedge C$.
Applying this to the inner expression with $A = P$, $B = Q$, and $C = R$:

$$[P \wedge (Q \wedge R)] \wedge S \equiv [(P \wedge Q) \wedge R] \wedge S$$

Now, applying the associative law a second time treating $A = (P \wedge Q)$, $B = R$, and $C = S$:

$$[(P \wedge Q) \wedge R] \wedge S \equiv (P \wedge Q) \wedge (R \wedge S)$$

15. **Truth Table Lines:**
There will be 2^n lines. Since each distinct letter has exactly 2 possible truth values (True or False), the total number of combinations for n variables is $2 \times 2 \times \cdots \times 2$ (n times).

16. **Formula from Truth Table:** The formula evaluates to False in exactly one case: when P is False and Q is True. In all other cases, it is True.
A logical statement that is true whenever P is true OR whenever Q is false matches this perfectly.
Therefore, the equivalent formula is: $P \vee \neg Q$.

17. **Formula from Truth Table:** This truth table represents the exclusive OR (XOR) operation, which evaluates to True if and only if exactly one of P or Q is True.
Using standard connectives, this means "P or Q, but not both P and Q".
The equivalent formula is: $(P \vee Q) \wedge \neg(P \wedge Q)$.

18. **Analyzing Argument Conclusions and Premises:**

- **If the conclusion is a tautology:** The argument is unconditionally valid. An argument is only invalid if it's possible for the premises to be true while the conclusion is false. Since a tautology is always true, this scenario is impossible.
- **If the conclusion is a contradiction:** The argument is valid if and only if the conjunction of all premises is also a contradiction. Otherwise, there would be a case where all premises are true but the conclusion is false (since a contradiction is always false).
- **If one of the premises is a tautology:** The tautology is always true and thus imposes no restrictions on the truth values. The validity of the argument depends entirely on the remaining premises.
- **If one of the premises is a contradiction:** The argument is vacuously valid. Because a contradiction is always false, it is impossible for all premises to be true simultaneously. Hence, you can never have the invalidating condition of "true premises and a false conclusion."